
# Sri Chaitanya IIT Academy., India.

AP, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON Central Office, Madhapur – Hyderabad

**Sec:** Sr-ICON-B-1 | **JEE-ADVANCE-2018-P1** | **Date: 09-05-21**

**Time: 07:00 AM to 10:00 AM** | **GTA-03** | **Max.Marks: 180**

## SYLLABUS:

**Physics** : TOTAL SYLLABUS

**Chemistry** : TOTAL SYLLABUS

**Mathematics** : TOTAL SYLLABUS

**Name of the Student:** ____________________ **H.T. NO:** | | | | | | | |

# JEE-ADVANCE-2018-P1-Model

**Time3h** **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

## PHYSICS:

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I(Q.N : 1 – 6) | Questions with Multiple Correct Choice (partial marking scheme) (+1,0) | +4 | -2 | 6 | 24 |
| Sec – II(Q.N : 7 – 14) | Questions with NV Integer Answer Type | +3 | 0 | 8 | 24 |
| Sec – II(Q.N : 15-18) | Questions with Comprehension Type (2 Comprehensions – 2 +2 = 4Q) | +3 | -1 | 4 | 12 |
| **Total** | | | | **18** | **60** |

## CHEMISTRY:

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I(Q.N : 19 – 24) | Questions with Multiple Correct Choice (partial marking scheme) (+1,0) | +4 | -2 | 6 | 24 |
| Sec – II(Q.N : 25 –32) | Questions with NV Integer Answer Type | +3 | 0 | 8 | 24 |
| Sec – II(Q.N : 33-36) | Questions with Comprehension Type (2 Comprehensions – 2 +2 = 4Q) | +3 | -1 | 4 | 12 |
| **Total** | | | | **18** | **60** |

## MATHEMATICS:

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I(Q.N:37 – 42) | Questions with Multiple Correct Choice (partial marking scheme) (+1,0) | +4 | -2 | 6 | 24 |
| Sec – II(Q.N :43–50) | Questions with NV Integer Answer Type | +3 | 0 | 8 | 24 |
| Sec – II(Q.N : 51-54) | Questions with Comprehension Type (2 Comprehensions – 2 +2 = 4Q) | +3 | -1 | 4 | 12 |
| **Total** | | | | **18** | **60** |

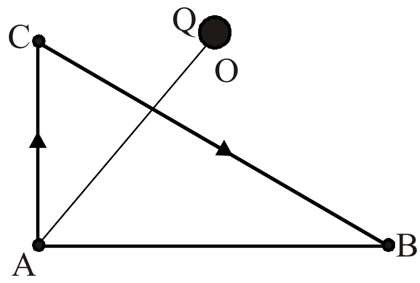
# PHYSICS

(Maximum Marks: 60)

## SECTION 1 (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question. Answer to each question will be evaluated according to the following marking scheme: Full Marks: +4 If only (all) the correction option(s) is (are) chosen. Partial Marks:+3 If all the four options are correct but ONLY three options are chosen. Partial Marks: +2 If three or more options are correct but ONY two options are chosen, both of which are correct options. Partial Marks:+1 If two or more options are correct but ONLY one options is chosen and it is a correct options. Zero Marks: 0 If none of the options is chosen(i.e. the question is unanswered) Negative Marks: -2 In all other cases. **For Example**: If first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option),without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

**1.** There is a fixed positive charge Q at O and A and B are points equidistant from O. A positive charge + q is taken slowly by an external agent from A to B along the line AC and then along the line CB.




**A)** The total work done on the charge is zero

**B)** The work done by the electrostatic force from A to C is negative

**C)** The work done by the electrostatic force from C to B is positive

**D)** The work done by electrostatic force in taking the charge from A to B is dependent on the actual path followed.

**2.** The $^{238}U$ nucleus has a binding energy of about 7.6 MeV per nucleon. If the nucleus were to fission into two equal fragments , each would have a kinetic energy of just over 100 MeV. From this , it can be concluded that

**A)** nuclei near A = 120 has mass less than half that of $^{238}U$.

**B)** nuclei near A = 120 have masses greater than half that of $^{238}U$

**C)** nuclei near A = 120 must be bound by about 6.7 MeV/ nucleon

**D)** nuclei near A = 120 must be bound by about 8.5 MeV/ nucleon

**3.** A string is under tension and stationary vibrations are produced in it. A, B, C, D are positions of consecutive nodes.

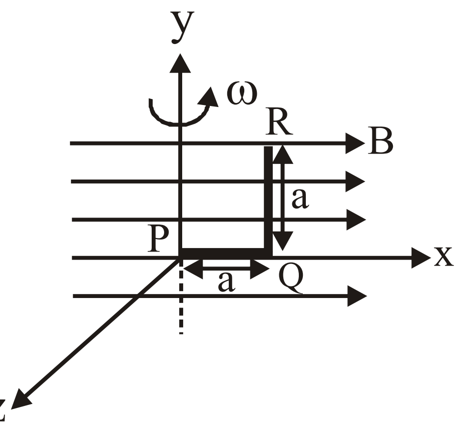
**A)** The distance AC = $\frac{\lambda}{2}$

**B)** The phase difference between a vibrating point on AB and another on BC is zero.

**C)** The amplitudes of vibration of different points between A and the midpoint of A and B are different

**D)** When every point of AB is having maximum displacement then every point of BC is having maximum displacement in opposite direction.

**4.** In a region there exist a magnetic field $B_0$ along positive x-axis. A metallic wire of length 2a and one side along x-axis and one side parallel of y-axis is rotating about y-axis with a angular velocity □. Then at the instant shown.




**A)** Potential difference across PQ is O

**B)** Potential difference across PQ is $\frac{1}{2}B_0\omega a^2$

**C)** Potential difference across QR is $\frac{1}{2}B_0\omega a^2$

**D)** Potential difference across QR is $B_0\omega a^2$

**5.** A sports car accelerates from zero to a certain speed in $t$ seconds. How long does it take for it to accelerate to twice that speed starting from rest, assuming the power of the engine to be constant (independent of velocity) and neglecting friction?

**A)** $\sqrt{2}\ t$ sec. **B)** 2t sec. **C)** 3t sec. **D)** 4t sec.

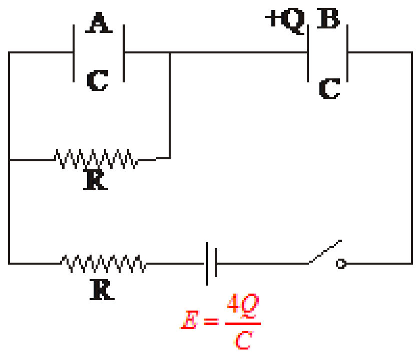
**6.** An open capillary tube is lowered in a vessel with mercury. The difference between the levels of the mercury in the vessel and in the capillary tube $\Delta h$ = 4.6mm. What is the radius of curvature of the mercury meniscus in the capillary tube? Surface tension of mercury is 0.46 N/m, density of mercury is 13.6 gm/cc.

**A)** $\frac{1}{340}m$ **B)** $\frac{1}{680}m$ **C)** $\frac{1}{1020}m$ **D)** Information insufficient

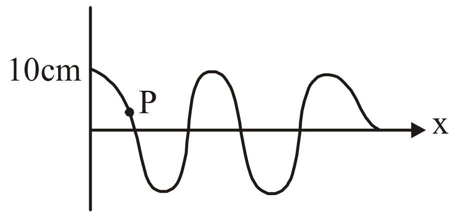
**SECTION 2 (Maximum Marks: 24)**

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) using the mouse and the on- screen virtual numeric keypad in the place designated to enter the answer. Answer to each question will be evaluated according to the following marking scheme: Full Marks :+3 if ONLY the correct numerical value is entered as answer. Zero Marks : 0 In all other cases

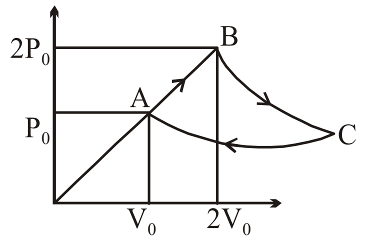
**7.** Consider the circuit shown in the figure. Capacitors A and B, each have capacitance C = 2F. The plates of capacitor A are shorted using a wire of resistance R = 1Ω while the left plate of capacitor B is given an initial charge Q = +4C. The switch is closed at a time t = 0. What will be the initial current (in ampere) drawn from the battery immediately after the switch is closed?



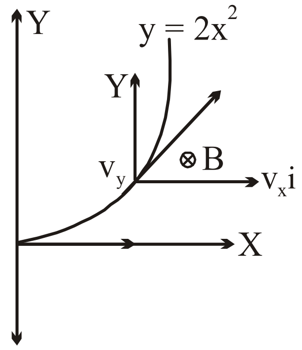
**8.** A transverse sinusoidal wave moves along a string in the positive x-direction at a speed of 10 cm/s. At a particular time t, the snap-shot of the wave is shown in figure. The speed of point P is $x\sqrt{2}\pi\, m/s$ when its displacement is $\frac{10}{\sqrt{2}}cm$, then the value of x is



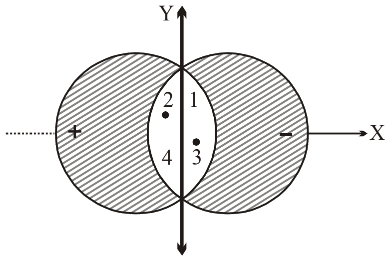

**9.** One mole of a diatomic gas is taken around a cyclic process shown in the figure. The process BC is adiabatic expansion and the process CA is isothermal compression, if the volume of the gas at C is given by $V_C = (2)^n V_0$, then the value of n is




**10.** A non-uniform magnetic field B varies with x. It exist for $x \ge 0$, into the xy plane. A charge q of mass m, enters the magnetic field at the origin with speed $v\hat{i}$. It is seen that it travels along $y = 2x^2$ curve. Find the value of B at x = 0. $\left(\text{Given} : \frac{mv}{q} = 2\right)$

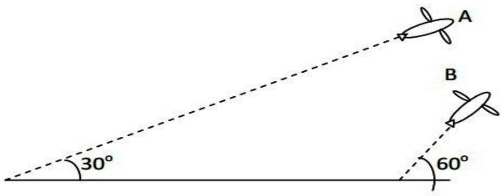


**11.** Two straight and very long nonmagnetic conductors insulated from each other carry a current in (+)ve and (–)ve Z-direction respectively. The cross sections of conductors are limited by circles of radius R in x-y plane with distance R between the centres.
If $B_{1y}$, $B_{2y}$, $B_{3y}$ and $B_{4y}$ are values of magnetic field along y-direction at points 1, 2, 3, and 4 respectively, then find the value of $\left(\frac{B_{1y}+B_{2y}+B_{3y}}{B_{4y}}\right)$. Assume current is uniformly distributed over cross section and $\mu_0 I = 100,\ R = 5$. (In SI units)




**12.** The work functions of Silver and Sodium are 4.6 and 2.3 eV, respectively. The ratio of the slope of the stopping potential versus frequency plot for Silver to that of Sodium is

**13.** Airplanes A and B are flying with constant velocity in the same vertical plane at angles $30^0$ and $60^0$ with respect to the horizontal respectively as shown in figure. The speed of A is $100\sqrt{3}\ ms^{-1}$. At time t = 0 s, an observer in A finds B at a distance of 500 m. This observer sees B moving with a constant velocity perpendicular to the line of motion of A. If at $t = t_0$, A just escapes being hit B, $t_0$ in second is




**14.** In a YDSE bi-chromatic light of wavelengths 400 nm and 560 nm are used. The distance between the slits is 0.1 mm and the distance between the plane of the slits and the screen is 1 m. The minimum distance between two successive regions of complete darkness is 4(2k+1) mm. The value of k is

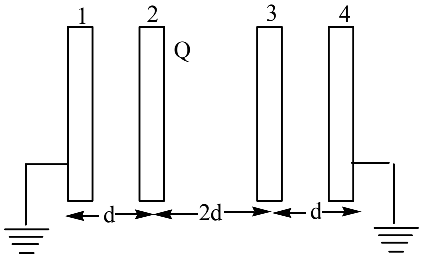
## SECTION 3 (Maximum Marks: 12)

This section contains **TWO (02)** paragraphs. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer. Answer to each question will be evaluated according to the following marking scheme: Full Marks :+3 If ONLY the correct options is chosen. Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered). *Negative Marks* : $-1$ In all other cases.

### PARAGRAPH "X"

(***There are two questions based on PARAGRAPH "X", the question given below is one of them***)

Four metallic plates are placed as shown in the fig. Plate 2 is given a charge 'Q' whereas all other plates are uncharged. Plates 1 and 4 are earthed. The area of each plate is same




**15.** The charge appearing on the right side of plate 3 is

**A)** zero **B)** $\frac{+Q}{4}$ **C)** $\frac{-3Q}{4}$ **D)** $\frac{Q}{2}$

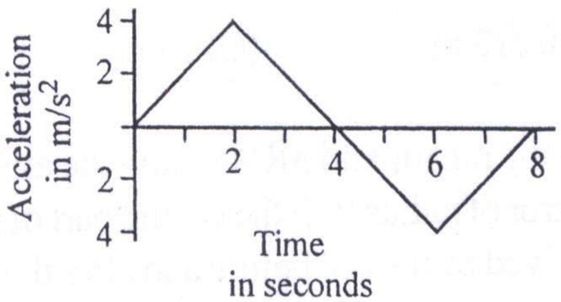
**16.** The charge appearing on the right side of plate 4 is

**A)** zero **B)** $\frac{+Q}{4}$ **C)** $\frac{-3Q}{4}$ **D)** $\frac{-Q}{2}$

### PARAGRAPH "A"

(***There are two questions based on PARAGRAPH "A", the question given below is one of them***)

A car starts from rest and accelerates as shown in the accompanying diagram.




**17.** At what time would the car be moving with the greatest velocity?

**A)** 2 seconds **B)** 4 seconds **C)** 6 seconds **D)** 8 seconds

**18.** At what time would the car be farthest from its original starting position?

**A)** 2 seconds **B)** 4 seconds **C)** 6 seconds **D)** 8 seconds

# CHEMISTRY

## SECTION 1 (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question. Answer to each question will be evaluated according to the following marking scheme: Full Marks: +4 If only (all) the correction option(s) is (are) chosen. Partial Marks:+3 If all the four options are correct but ONLY three options are chosen. Partial Marks: +2 If three or more options are correct but ONY two options are chosen, both of which are correct options. Partial Marks:+1 If two or more options are correct but ONLY one options is chosen and it is a correct options. Zero Marks: 0 If none of the options is chosen(i.e. the question is unanswered) Negative Marks: -2 In all other cases. **For Example**: If first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option),without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

**19.** Which of the following are correct about Tyndall effect?

**A)** True solution do not show Tyndall effect due to very small size of the particles.

**B)** The diameter of the particles of the dispersed phase must not be much smaller than the wavelength of light used.

**C)** Tyndall effect is very weak in case of lyophobic sols.

**D)** The refractive index of the dispersed phase and dispersion medium must differ considerably.

**20.** Which of the following is/are false ?

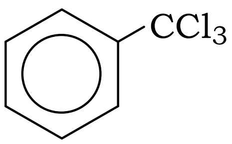
**A)** 1 M NaCl solution has higher freezing point than 1 M glucose solution

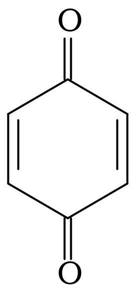
**B)** 1 M glucose solution has same boiling point as 1 M sucrose solution

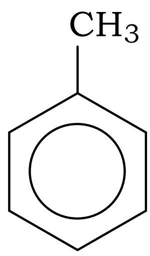
**C)** Molecular weight of benzoic acid in benzene will be double than expected

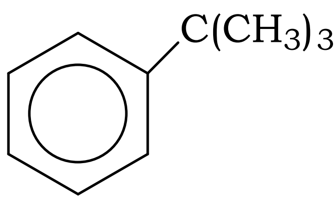
**D)** Van't Hoff factor (i) > 1 if solute undergoes association

**21.** Which of the following compounds show hyper conjugation

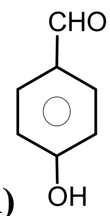
**A)** 

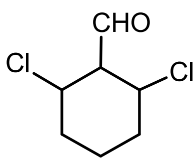

**B)** 

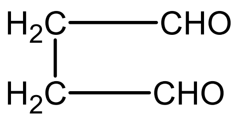

**C)** 

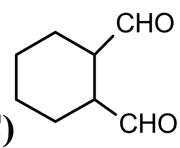

**D)** 


**22.** Which of the following compounds do not give Cannizaro reaction ?

**A)** 


**B)** 


**C)** 


**C)** 

**23.** Which of the following oxides dissolve in sodium hydroxide forming corresponding salts.

**A)** $Al_2O_3$ **B)** $ZnO$ **C)** $SnO_2$ **D)** $PbO_2$

**24.** The borax bead test can be used to detect the presence of

**A)** $Al^{+3}$ **B)** $Cr^{+3}$ **C)** $Cu^{2+}$ **D)** $Co^{2+}$

## SECTION 2 (Maximum Marks: 24)

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**25.** Molar conductance of saturated $BaSO_4$ solution is 400 $ohm^{-1}$ $cm^2$ $mol^{-1}$ and its specific conductance is $4 \times 10^{-5}$ $ohm^{-1}$ $cm^{-1}$. Hence solubility product value of $BaSO_4$ can be concluded as $x \times 10^{-8}$ $M^2$. Identify x.

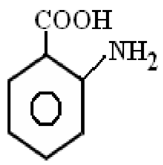
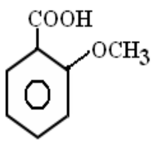
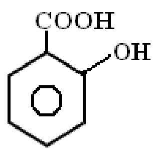
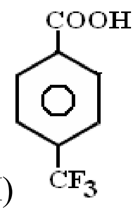
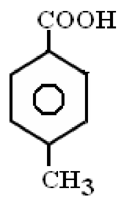
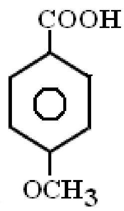
**26.** Half - life period of the decomposition of a compound is 20 minutes. If the initial concentration is doubled, the half - life period is reduced to 10 mins. What is the order of the reaction?

**27.** Solubility of $Pb(OH)_2(s)$ in pure water is $10^{-4}M$. How many times the solubility of $Pb(OH)_2$ will increase in buffer of $pH = 10$?

**28.** For a gas obeying $P(V-b) = RT$ a graph is plotted between Volume (V lit.) on Y-axis and Temperature (K) on X-axis at a constant pressure of 0.04105 atm. Find the slope of the line?

**29.** How many of the following acids are more acidic than benzoic acid?

I) HCOOH, II) $CH_3 - C \equiv C - COOH$, III) $CF_3 - COOH$,

IV)  , V)  , VI)  , VII)  ,VIII)  , IX) 

**30.** How many of the following reagents can be used to distinguish between hex-1-yne and hex-2-yne?

(a) $CuCl / NH_3$ (b) $AgNO_3 / NH_3$ (c) Na metal

**31.** How many oxygen atoms of $[SiO_4]^{4-}$ are shared in three dimensional sheet silicate ?

**32.** Number of copper (atomic weight 64) atoms present in a coin made up of bell metal is $3.0115 \times 10^{22}$. The weight of the coin in grams is ____

## SECTION 3 (Maximum Marks: 12)

This section contains **TWO (02)** paragraphs. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer. Answer to each question will be evaluated according to the following marking scheme: Full Marks :+3 If ONLY the correct options is chosen. Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered). *Negative Marks* : $-1$ In all other cases.

### PARAGRAPH "X"

**(*There are two questions based on PARAGRAPH "X", the question given below is one of them*)**

Chlorine forms a number of oxides such as $Cl_2O, ClO_2, Cl_2O_6$ and $Cl_2O_7$. All of them are unstable and highly reactive. Chlorine forms also oxyacids such as $HClO, HClO_2, HClO_3$ and $HClO_4$.

**33.** Which of the following oxyacids of chlorine is most acidic ?

A) $HClO_3$ B) $HClO_2$ C) $HClO_4$ D) $HClO$

**34.** Which of the following gives $Cl_2O_7$ when treated with $P_2O_5$.

A) $HClO_4$ B) $HClO_3$ C) $HClO$ D) $HClO_2$

### PARAGRAPH "A"

**(*There are two questions based on PARAGRAPH "A", the question given below is one of them*)**

$$(A)\xrightarrow[250^0C]{Al_2O_3}(B)\xrightarrow[(ii)\,AgOH]{HI}(C)\xrightarrow[150^0C]{Al_2O_3}(B)\xrightarrow[(ii)\,H_2O_2,OH^-]{(i)B_2H_6}(A)$$

In the above reaction sequence (A) and (C) are isomers. Molecular formula of B is $C_5H_{10}$, which can also be obtained from the product of the reaction with $CH_3CH_2MgBr$ and $(CH_3)_2CO$ and followed by acidification and heating.

**35.** Identify structure of A

**A)** $CH_3-CH_2-CH_2-\underset{\underset{OH}{|}}{CH}-CH_3$

**B)** $(CH_3)_2CHCHOHCH_3$

**C)** $(CH_3)_3C-CH_2-OH$

**D)** $CH_3-CH_2-\underset{\underset{OH}{|}}{CH}-CH_2-CH_3$

**36.** Identify the structure of B

**A)** $(CH_3)_2-CH-CH=CH_2$

**B)** $CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-C=CH_2$

**C)** $(CH_3)_2C=CHCH_3$

**D)** $CH_2=CH-\underset{\underset{CH_3}{|}}{CH}-CH_3$

# MATHEMATICS

## SECTION 1 (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question. Answer to each question will be evaluated according to the following marking scheme: Full Marks: +4 If only (all) the correction option(s) is (are) chosen. Partial Marks:+3 If all the four options are correct but ONLY three options are chosen. Partial Marks: +2 If three or more options are correct but ONY two options are chosen, both of which are correct options. Partial Marks:+1 If two or more options are correct but ONLY one options is chosen and it is a correct options. Zero Marks: 0 If none of the options is chosen(i.e. the question is unanswered) Negative Marks: -2 In all other cases. **For Example**: If first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option),without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

**37.** Given that system of equations $2x-y+z=0, x-2y+z=0, tx-y+2z=0$ has infinitely many solutions. If f(x) is a continuous function such that $f(5+x)+f(x)=2$ *for all* $x\in R$ and value of $\int_0^{-2t} f(x)dx$ is $\mu$ then

**A)** t = 5 **B)** period of f (x) is 5

**C)** $\mu = 2t$ **D)** $\mu = -2t$

**38.** For every pair of function $f,g:[0,1]\to R$ such that $\max\{f(x):x\in[0,1]\}=\max\{g(x):x\in[0,1]\}$. The correct statement is /are

**A)** $(f(c))^2+3f(c)=(g(c))^2+3g(c)$ *for some* $c\in[0,1]$

**B)** $(f(c))^2+f(c)=(g(c))^2+3g(c)$ *for some* $c\in[0,1]$

**C)** $(f(c))^2+3f(c)=(g(c))^2+g(c)$ *for some* $c\in[0,1]$

**D)** $(f(c))^2=g(c)^2$ *for some* $c\in[0,1]$

**39.** A circle concentric to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a>b$ also passes through the foci $S_1,S_2$. If circle and ellipse intersect at P in first quadrant then which of the following is/are true.

**A)** ordinate of P is $\frac{a^2}{be}$ **B)** ordinate of P is $\frac{b^2}{ae}$

**C)** Triangle $PS_1S_2$ area in $b^2$ **D)**triangle $PS_1S_2$ are in $a^2$

**40.** Let the function $g:(-\infty,\infty)\to\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$ be given as $g(u)=2\tan^{-1}(e^u)-\frac{\pi}{2}$. Then g is

**A)** odd function **B)** decreasing function

**C)** one one function **D)** onto function

**41.** If $f^{11}(x)<0$ *for all* $x\in R$ *and* $g(x)=f(x^2-2)+f(6-x^2)$ then $g(x)$ is increasing in the intervals

**A)** $[0,2]$ **B)** $[-2,0)$ **C)** $[2,\infty)$ **D)** $(-\infty,-2]$

**42.** A triangle has vertices at $(1,1,1),(2,2,2),(1,1,y)$ and area of the triangle is cosec $\frac{\pi}{4}$ square units then value of y is /are

**A) -3** **B) -1** **C) 0** **D) 3**

**SECTION 2 (Maximum Marks: 24)**

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) using the mouse and the on- screen virtual numeric keypad in the place designated to enter the answer. Answer to each question will be evaluated according to the following marking scheme: Full Marks :+3 if ONLY the correct numerical value is entered as answer. Zero Marks : 0 In all other cases

**43.** Given $S=\{1,2,3,4,5,6,7\}$. The number of subsets A of S such that, if $x\in A$ then $x+1\notin A$, is ____

**44.** Area bounded by the curves $y=\ln x, y=\sin^4\pi x$ between origin and first point of intersection of the two curves, is K then 2K_____

**45.** If $[x]$ is greatest integer function, then $\lim\limits_{x\to\infty}\frac{\log(x+2)+[x]+2}{\left[x+\sum_{r=0}^{\infty}\left(\frac{1}{2}\right)^r\right]}=$ ___

**46.** Six points $(x_i,y_i), i=1,2,3,4,5,6$ are on the circle $x^2+y^2=4$ such that $\sum_{i=1}^{6}x_i=8, \sum_{i=1}^{6}y_i=4$. The ortho centre of triangle formed by first three points is A, centroid of triangle formed by remaining three points is B. If line AB passes through a fixed point (h,k) then value of $\frac{k}{h}$ is ___

**47.** Given $X=\{1,2,3,.....,25\}$. A student formed a function from X onto X. The probability that function having primes onto primes is $\frac{1}{25(a.b.cd)}$ where a,b,c,d are primes then $a+b+c+d=$_______

**48.** If coefficient of $x^{10}$ in the expansion of $(1+x^2-x^3)^8$ is K then value of $\frac{K}{10}$ is ____

**49.** The position vectors of four angular points of a tetrahedron are $A(j+2k), B(3i+k), C(4i+3j+6k), D(2i+3j+2k)$. If volume of tetrahedron is V and distance between the lines AB and CD is $l$ then value of $\frac{l^2}{v+4}$ is ______

**50.** If $f(x)$ is a polynomial function satisfying $f(x)+f(1-x)=1$ *then* $\int_{1/4}^{3/4} f(f(x))dx =$____

**SECTION 3 (Maximum Marks: 12)**

This section contains **TWO (02)** paragraphs. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer. Answer to each question will be evaluated according to the following marking scheme: Full Marks :+3 If ONLY the correct options is chosen. Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered). *Negative Marks* : $-1$ In all other cases.

**PARAGRAPH "X"**

(***There are two questions based on PARAGRAPH* "X", *the question given below is one of them***)

Let $a_1, a_2, ......a_n$ be n increasing numbers in A.P and $S_n$ be the sum of first n terms of that A.P. If the point $(n, S_n)$ lies on a parabola with vertex (-2,-1) for all $n \in N$

**51.** Length of latus rectum of that parabola, is

**A)** $\frac{7}{2}$ **B)** 8 **C)** 4 **D)** 1

**52.** $y = 2x + c$ is tangent to that parabola when the value of c is

**A)** -1 **B)** 1 **C)** 4 **D)** $\frac{7}{2}$

**PARAGRAPH "A"**

(***There are two questions based on PARAGRAPH* "A", *the question given below is one of them***)

Given $f(z) = z^{12} + 2z^{11} + 3z^{10} + ... + 12z + 13$ *and* $\alpha = \cos\frac{2\pi}{13} + i\sin\frac{2\pi}{13}$

**53.** Value of $f(\alpha).f(\alpha^2).....f(\alpha^{12})$ is equal to

**A)** $13^{11}$ **B)** $13^{12}$ **C)** $12^{13}$ **D)** $12^{11}$

**54.** If $\text{Re}\left(f(z) - \frac{z(z^{13}-1)}{(z-1)^2}\right)$ *is* $\frac{13}{4}$ then locus of z is

**A)** circle with centre (0,0) **B)** parabola with vertex (0,0)

**C)** Parabola with vertex (-1,0) **D)** Circle with centre (-1,0)